CYBERSECURITY ADVANCED FRAMEWORK

Implementing Great Cybersecurity Practices

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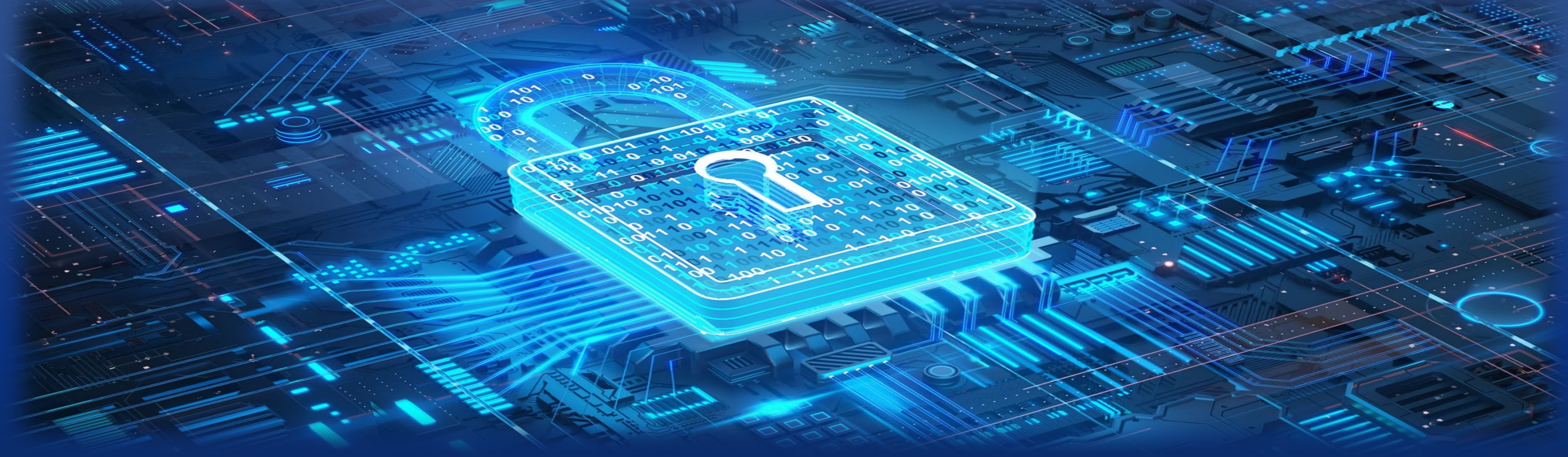


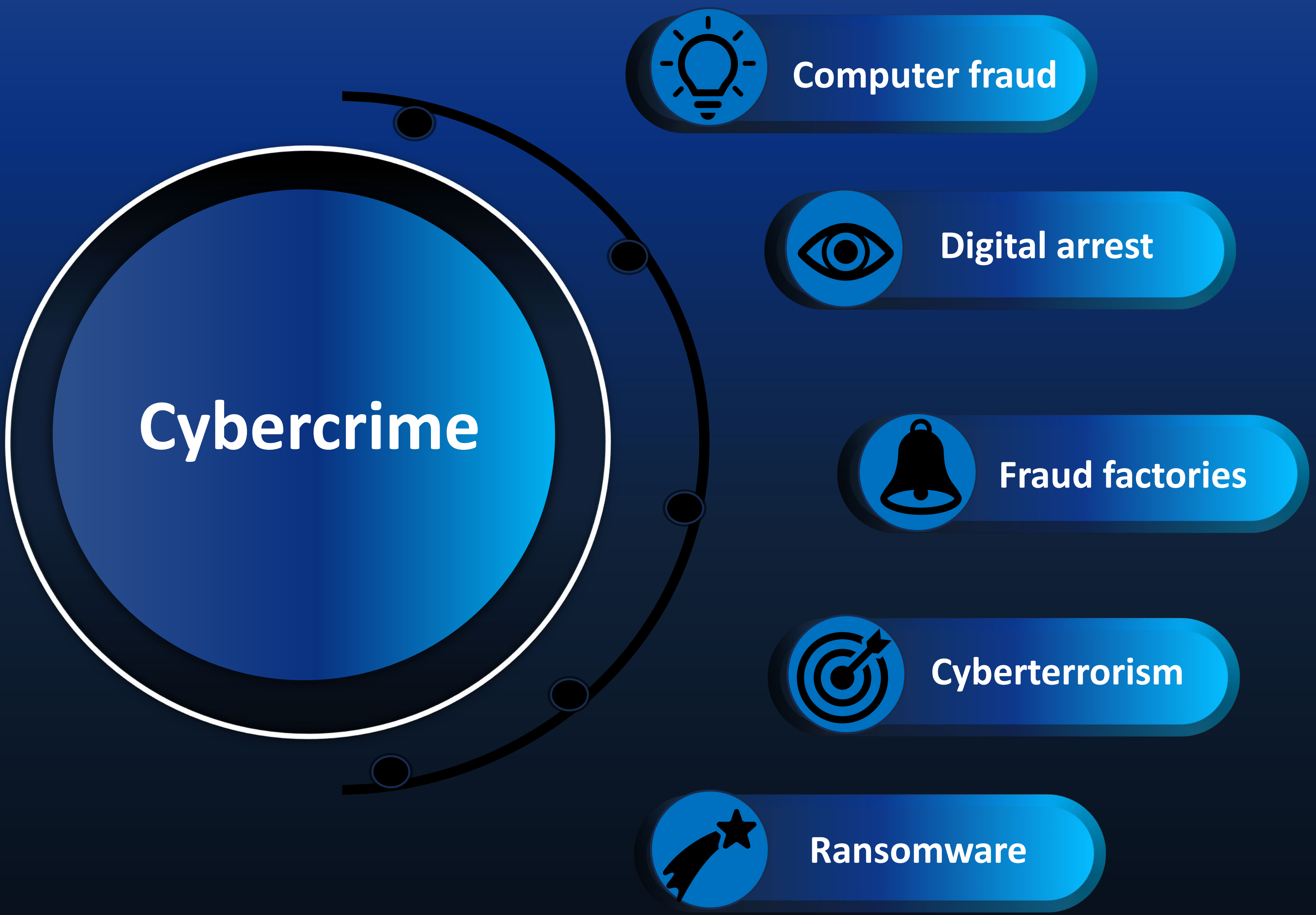
Why Computer security is important

Computer security is the digital equivalent of locking your front door and hiding your valuables in a safe. In an era where our lives—from bank accounts and medical records to private conversations and career projects—are stored online, the stakes are incredibly high. Without robust security, you are vulnerable to **identity theft, financial fraud, and data breaches** that can take years to resolve.

- **Database protection**
- **Identity and Access Management (IAM)**

Cybercrime

Cybercrime encompasses a wide range of criminal activities that are carried out using digital devices and/or networks. It has been variously defined as "a crime committed on a computer network, especially the Internet.



Security exploits

Brute-force attack

Password cracking

Packet analyzer

Spoofing attack

To protect against security exploits, you need a layered defense that covers your network, individual devices (endpoints), and application code. Exploits target known or unknown vulnerabilities to gain unauthorized access or execute malicious code.

Integrated Security Operations

Incident Response & Containment

Automated playbooks enable immediate isolation of compromised systems, preventing lateral movement and data exfiltration.

Continuous Monitoring & Optimization

24/7 security operations center with dedicated analysts monitoring endpoints, networks, cloud environments, and applications. Quarterly threat assessments, penetration testing, and security posture reviews ensure sustained protection against emerging attack vectors and compliance with regulatory requirements.

Threat Detection & Analysis

Advanced SIEM platforms with machine learning algorithms analyze millions of security events daily. Real-time behavioral analytics identify anomalous patterns, zero-day exploits, and advanced persistent threats before they compromise critical systems.

Risk Assessment & Vulnerability Management

Proactive identification and remediation of security vulnerabilities is critical to maintaining a strong security posture. Our comprehensive assessment framework combines automated scanning, manual penetration testing, and continuous monitoring to identify and prioritize risks before they can be exploited.

Vulnerability Scanning & Assessment

Automated tools continuously scan endpoints, networks, and applications for known vulnerabilities and misconfigurations. Our assessment covers CVSS scoring, exploitability analysis, and business impact evaluation. Weekly scans with monthly deep-dive assessments ensure comprehensive coverage across all infrastructure components.

Penetration Testing & Red Team Operations

Quarterly authorized penetration tests simulate real-world attack scenarios, testing defenses against advanced threat actors. Our red team exercises validate security controls, identify attack chains, and assess organizational response capabilities. Results drive prioritized remediation roadmaps with executive-level risk reporting.

Enterprise Security Infrastructure

Advanced Threat Prevention

Deploy next-generation firewalls, intrusion prevention systems, and endpoint detection and response (EDR) solutions.

Comprehensive Logging & Analytics

Centralize security logs from all infrastructure components into a unified SIEM platform..

Continuous Security Training & Awareness

Establish mandatory security awareness programs for all employees with role-specific training for IT and security personnel

Automation & OrchestSecurityration

Implement SOAR platforms to automate routine security tasks, reducing response times from hours to minutes.

Capabilities and access control lists

Within computer systems, two of the main security models capable of enforcing privilege separation are access control list(ACLs) and role-based access control (RBAC).

- An access-control list (ACL), with respect to a computer file system, is a list of permissions associated with an object.

- Role-based access control is an approach to restricting system access to authorized users, used by the majority of enterprises with more than 500 employees,and can implement mandatory access control (MAC).

Technology Stack Overview

Endpoint Protection

- EDR/XDR platforms
- Antivirus with AI detection
- Disk encryption
- Application control

Cloud Security

- CSPM tools
- Cloud-native security
- Configuration management
- API security gateways

Network Security

- NGFW with SSL inspection
- IPS/IDS systems
- DNS filtering
- VPN concentrators

Core Security Framework Components

Zero Trust Architecture

Never trust, always verify. Micro-segmentation and continuous authentication across all resources regardless of location.

Network Defense

Next-generation firewalls, encrypted traffic analysis, and SD-WAN security controls to protect data in transit.

Identity Management

Multi-factor authentication, single sign-on, and privileged access management to secure user credentials.

Threat Detection

SIEM platforms with behavioral analytics, anomaly detection, and automated incident response capabilities.

Critical Benchmarks and Targets

1-10-60

Rule

The gold standard for security operations: 1 minute to detect, 10 minutes to analyze, and 60 minutes to remediate.

60-75%

Analyst Utilization

Number that indicates the Optimal SOC team workload.

33% - 5%.

Training Effectiveness

The number of phishing failures that can be reduced with the security training.

99.9%

Policy Compliance

Which will target employee acknowledgment of security policies.

Implementation Roadmap

Assessment & Planning

Comprehensive security audit identifying vulnerabilities, compliance gaps, and risk prioritization across all systems.

Architecture Design

Develop security architecture blueprint integrating existing infrastructure with new controls and automation workflows.

Technology Deployment

Phased rollout of security solutions with continuous monitoring, testing, and optimization based on real-world performance.

Continuous Improvement

Regular penetration testing, threat intelligence integration, and security posture assessments ensuring sustained protection.

Scan me!

THANKS TO ALL!